



Integration Challenges in Operation

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Objective

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At the end of this lesson, students should be able to identify common integration challenges in organizations, explain how legacy systems hinder modernization initiatives, describe how data silos negatively affect decision-making processes, analyze interoperability issues in multi-system environments, and understand the governance and compliance considerations involved in system integration.

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Why Integration is Challenging?



As organizations grow, their information systems also expand. This growth creates significant integration challenges.

As organizations grow, they acquire:

- ❑ New software
- ❑ Different platforms
- ❑ Multiple databases
- ❑ Department-specific systems

This growth creates significant integration challenges.

Challenges in Integrating Heterogeneous Systems

System Integration Challenges

1. Legacy Systems

- Older systems were not designed for integration
- Limited or no APIs
- Risk of disrupting operations

2. Data Incompatibility

- Different data formats
- Inconsistent data definitions
- Data quality issues

3. Security Risks

- Multiple access points
- Weak authentication between systems
- Data exposure during transmission

4. Scalability and Performance

- Integration may slow down systems
- High transaction volumes stress interfaces

5. Cost and Complexity

- Custom integrations are expensive
- Maintenance increases over time



Major Integration Challenges



1 Legacy Systems

- ❑ Older systems were not designed for integration
- ❑ Limited or no APIs
- ❑ Risk of disrupting operations

They may:

- ❖ Use outdated programming languages
- ❖ Run on old hardware
- ❖ Lack documentation
- ❖ Be difficult to modify

⚠ Integration Challenges

- No API support
- Incompatible data formats
- High maintenance cost
- Security vulnerabilities
- Risk of system failure during migration

Major Integration Challenges

2 Data Silos

A **data silo** occurs when data is isolated within one department or system and is not shared across the organization.

Impact of Data Silos

- ☐ Duplicate data
- ☐ Inconsistent information
- ☐ Poor decision-making
- ☐ Delayed reporting
- ☐ Lack of enterprise visibility



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Major Integration Challenges



3 Interoperability Issues

Interoperability refers to the ability of different systems to communicate and exchange data effectively.

Interoperability issues arise when systems:

- ☐ Use different data formats
- ☐ Use incompatible technologies
- ☐ Follow different standards

Types of Interoperability

1. **Technical Interoperability** – Network and protocol compatibility
2. **Semantic Interoperability** – Shared meaning of data
3. **Organizational Interoperability** – Aligned policies and processes



Major Integration Challenges



④ Governance and Compliance Considerations

Governance refers to policies and decision-making structures that guide system management.

Compliance refers to adherence to laws, regulations, and organizational standards.

Why Governance Matters in Integration

Integration affects:

- ❑ Data access
- ❑ Data ownership
- ❑ Security control
- ❑ Audit tracking

Without governance:

- ❖ Unauthorized integrations may occur
- ❖ Security gaps may arise
- ❖ Data misuse becomes possible

Major Integration Challenges



4 Governance and Compliance Considerations

Compliance Considerations

Organizations must comply with:

- ☐ Data privacy laws
- ☐ Financial regulations
- ☐ Industry standards
- ☐ Security frameworks

Governance Challenges

- Who owns the data?
- Who approves integrations?
- Who manages APIs?
- Who ensures compliance?



Summary of Integration Challenges:

Challenge	Impact	Risk Level
Legacy Systems	Hard to modernize	High
Data Silos	Poor visibility	High
Interoperability Issues	System incompatibility	Medium–High
Governance Gaps	Security & compliance risks	Very High

Thank
You